W1 – Introduction to Vectors Unit 4

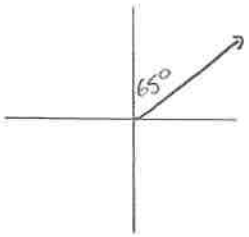
MCV4U

Iensen

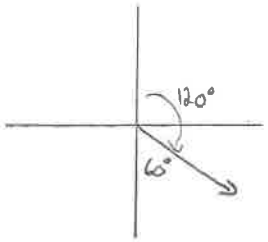
SOLUTIONS

1) Convert each true bearing to its equivalent quadrant bearing.

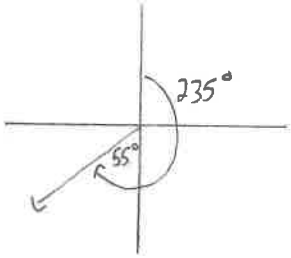
a) 065° = N65°E



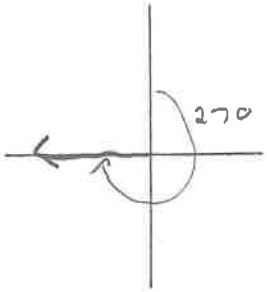
b) 120° = S60°E



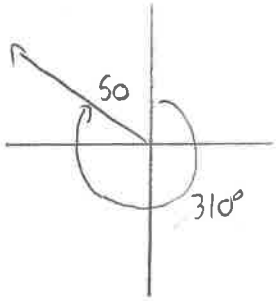
c) 235° = S55°W



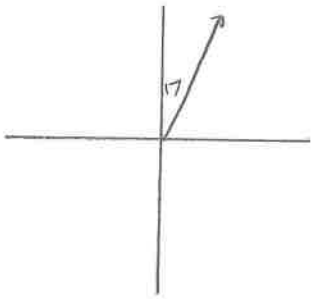
d) 270° = W



e) 310° = N50°W

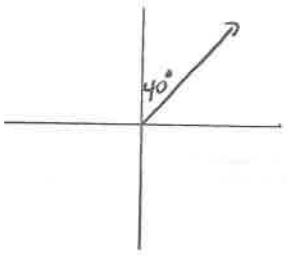


f) 017° = N17°E

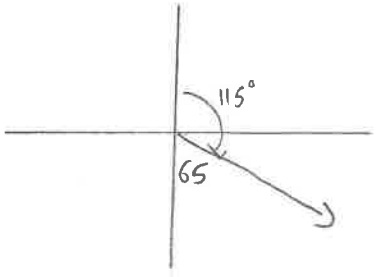


2) Convert each quadrant bearing to its equivalent true bearing.

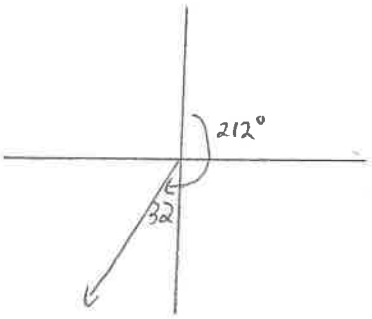
a) N40°E = 040°



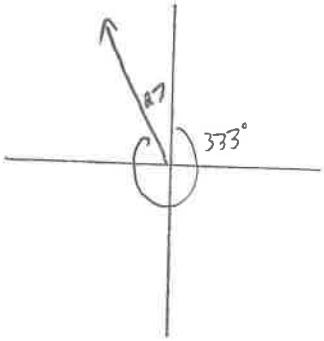
b) S65°E = 115°



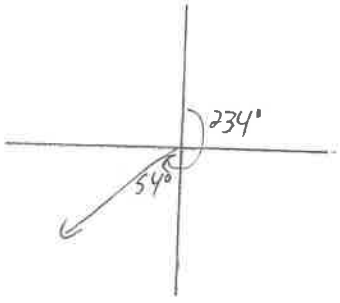
c) S32°W = 212°



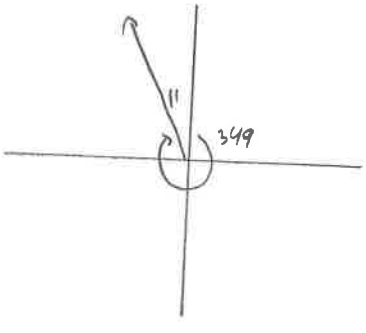
d) N27°W = 333°



e) S54°W = 234°



f) N11W = 349°



3) In the space to the right, draw and name...

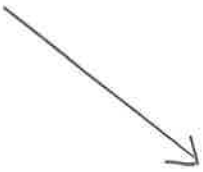
a) a vector parallel to $\overrightarrow{AB}$

b) a vector opposite to $\overrightarrow{AB}$

c) a vector equivalent to $\overrightarrow{AB}$

a)

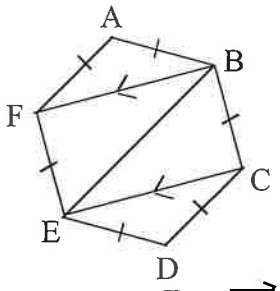
b)



c)

4) Name all the equivalent vectors in each diagram.

a)



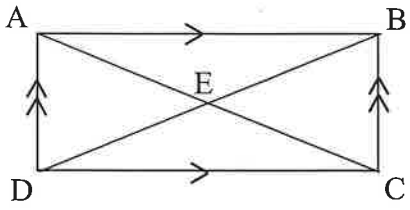
$\overrightarrow{AB} = \overrightarrow{ED}$ $\overrightarrow{BA} = \overrightarrow{DE}$

$\overrightarrow{AF} = \overrightarrow{CD}$ $\overrightarrow{FA} = \overrightarrow{DC}$

$\overrightarrow{BC} = \overrightarrow{FE}$ $\overrightarrow{CB} = \overrightarrow{EF}$

$\overrightarrow{CE} = \overrightarrow{BF}$ $\overrightarrow{EC} = \overrightarrow{FB}$

b)



$\overrightarrow{AD} = \overrightarrow{BC}$ $\overrightarrow{DA} = \overrightarrow{CB}$

$\overrightarrow{AE} = \overrightarrow{EC}$ $\overrightarrow{CE} = \overrightarrow{EA}$

$\overrightarrow{DE} = \overrightarrow{EB}$ $\overrightarrow{BE} = \overrightarrow{ED}$

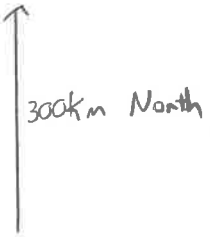
$\overrightarrow{AB} = \overrightarrow{DC}$ $\overrightarrow{BA} = \overrightarrow{CD}$

5) State the opposite of each vector.

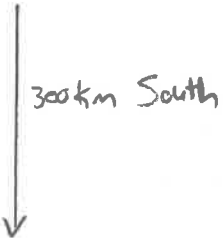
a) 300 km north

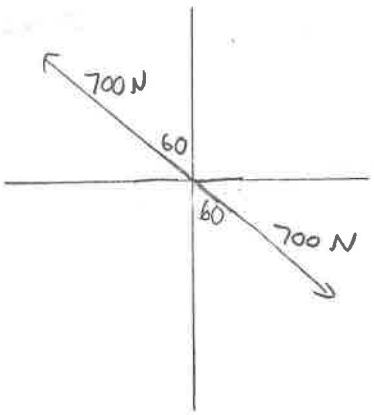
b) 700 N on a bearing of 120°

c) 180 km/h on a quadrant bearing of S25°W

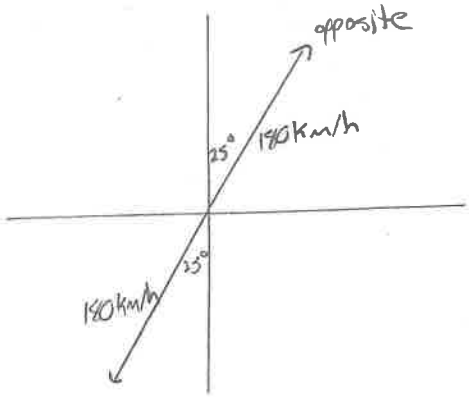


opposite:





opposite: 700 N on a bearing of 300°

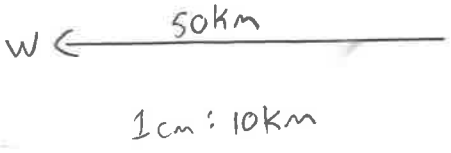


opposite:

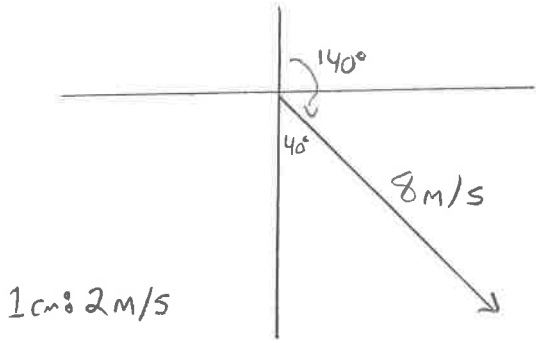
180 km/h on a quadrant bearing of N25°E.

6) Use an appropriate scale to draw each vector. Label magnitude, direction

a) displacement of 50 km west

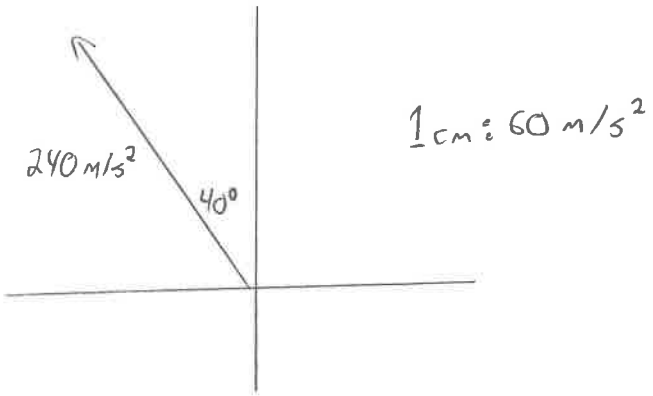


b) velocity of 8 m/s on a true bearing of 140°

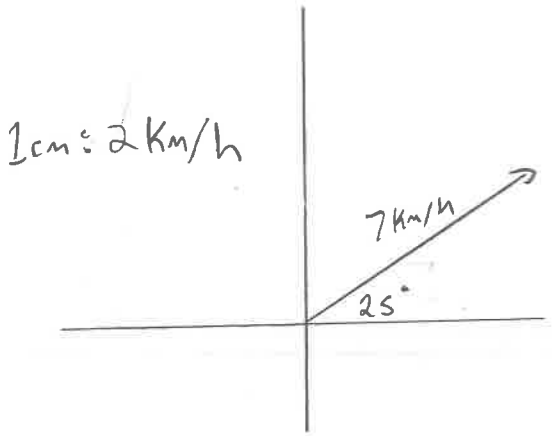


c) force of 1200 N downward

d) acceleration of 240 m/s^2 on a quadrant bearing of N40°W



e) velocity of 7 km/h at 25° to the horizontal



7) State whether the following are vectors or scalars:

a) A table weighs 80 N VECTOR

b) A woman's age is 60 years old SCALAR

)0 km/h SCALAR

d) An elevator lifts a person 20 m VECTOR

8)a) Which vectors are parallel to $\overrightarrow{AB}$?

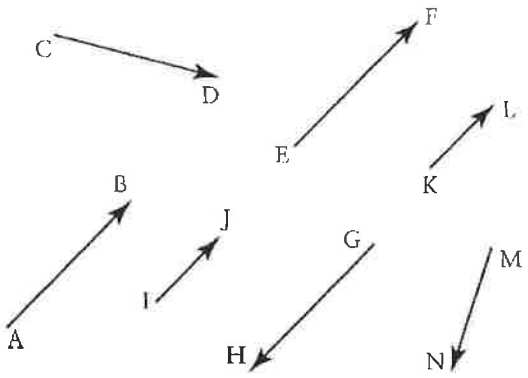
$\overrightarrow{IJ}$, $\overrightarrow{GH}$, $\overrightarrow{KL}$, $\overrightarrow{EF}$

b) Which vectors are equivalent to $\overrightarrow{AB}$?

$\overrightarrow{EF}$

c) Which vectors are opposite to $\overrightarrow{AB}$?

$\overrightarrow{GH}$

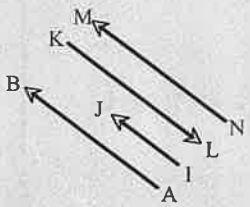


Answer Key:

1) a) N65°E **b)** S60°E **c)** S55°W **d)** W **e)** N50°W **f)** N17°E

2) a) 40° **b)** 115° **c)** 212° **d)** 333° **e)** 234° **f)** 349°

3) Diagrams may vary. For example, in the diagram shown, $\overrightarrow{IJ}$ is parallel to $\overrightarrow{AB}$, $\overrightarrow{KL}$ is opposite to $\overrightarrow{AB}$, and $\overrightarrow{MN}$ is equivalent to $\overrightarrow{AB}$.



4) a) $\overrightarrow{AB} = \overrightarrow{ED}$, $\overrightarrow{BC} = \overrightarrow{FE}$, $\overrightarrow{CD} = \overrightarrow{AF}$, $\overrightarrow{DE} = \overrightarrow{BA}$, $\overrightarrow{EF} = \overrightarrow{CB}$, $\overrightarrow{FA} = \overrightarrow{DC}$, $\overrightarrow{FB} = \overrightarrow{EC}$, $\overrightarrow{BF} = \overrightarrow{CE}$

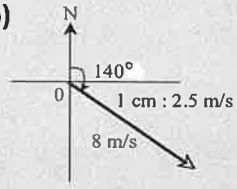
b) $\overrightarrow{AB} = \overrightarrow{DC}$, $\overrightarrow{BA} = \overrightarrow{CD}$, $\overrightarrow{AD} = \overrightarrow{BC}$, $\overrightarrow{DA} = \overrightarrow{CB}$, $\overrightarrow{DE} = \overrightarrow{EB}$, $\overrightarrow{BE} = \overrightarrow{ED}$, $\overrightarrow{AE} = \overrightarrow{EC}$, $\overrightarrow{CE} = \overrightarrow{EA}$

5) a) 300 km south **b)** 700 N on a bearing of 300° **c)** 180 km/h on a quadrant bearing of N25°E

6) Diagrams may vary.

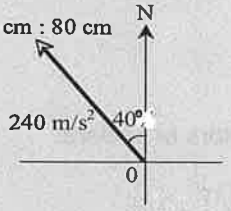
a)

b)

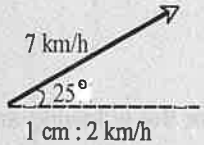


c)

d)



e)



7) a) Vector – weight is due to the force of gravity and therefore has a direction

b) Scalar – this has no direction **c)** Scalar – there is no direction so this is just speed

d) Vector – there is magnitude and direction (up)

8)a) $\overrightarrow{EF}, \overrightarrow{IJ}, \overrightarrow{KL}, \overrightarrow{GH}$ **b)** $\overrightarrow{EF}$ **c)** $\overrightarrow{GH}$